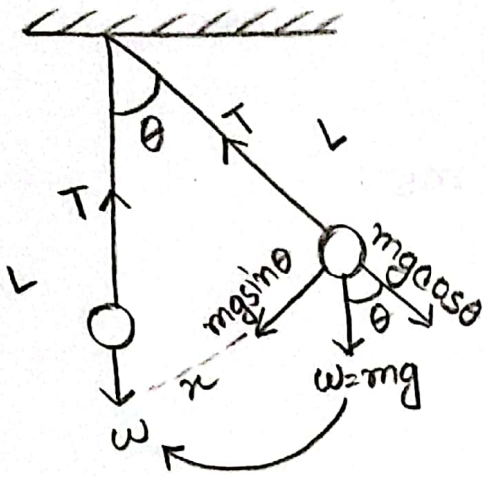


θ হৈলিকো কোণ ঙ্গেদর করে সে বরাবর $\cos\theta$ হবে।

Tension Force: Tension is defined as the force transmitted through a rope, string or wire when pulled by forces acting from opposite sides.

☞ Tension always thread (খুঁটা) বরাবর কাজ করবে।

Tension force এবং $mg \cos\theta$ একে অপরের সম্পূর্ণ বিপরীত দিকে কাজ করে, তাই cancel হয়ে যাবে, সুতরাং $mg \sin\theta$ থাকবে,

$T = mg \cos\theta$ বিপরীত এবং সমান।

$$\text{Restoring} = -mg \sin\theta$$

$$= -mg\theta$$

$$x = L\theta$$

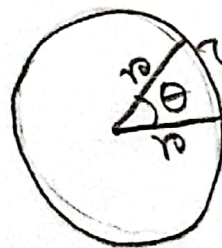
$$\Rightarrow \frac{d^2x}{dt^2} = L \frac{d^2\theta}{dt^2} \quad [\text{এর করার জন্য দুইবার differentiation}]$$

$$\Rightarrow a = L \frac{d^2\theta}{dt^2}$$

θ এর মান যদি অনেক ছোট হয় তখন $\sin\theta \approx \theta$

θ = angular displacement

x = linear displacement



$$\therefore s = r\theta$$

$$\therefore x = L\theta$$

We know,

$$F = -mgsin\theta$$

in spring, $\omega^2 = \frac{k}{m}$

$$\Rightarrow ma = -mgsin\theta$$

in angle, $\omega^2 = \frac{g}{L}$

$$\Rightarrow ma = -mg\theta$$

$$\Rightarrow a = -g\theta$$

in angular displacement,

$$\Rightarrow L \frac{d^2\theta}{dt^2} = -g\theta$$

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{g}{L}\theta$$

in linear displacement,

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\omega^2\theta$$

$$\frac{d^2x}{dt^2} + \omega^2x = 0$$

$$\Rightarrow \boxed{\frac{d^2\theta}{dt^2} + \omega^2\theta = 0}$$

now, $\omega^2 = \frac{g}{L}$

$$\Rightarrow \omega = \sqrt{\frac{g}{L}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\frac{g}{L}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{L}{g}}$$

(proved)